Natural Language Processing

Autumn 2026

Week 3: Part-of-Speech Tagging

Assignment 1 Questions?

ANY QUESTIONS ABOUT ASSIGNMENT 1?

Part-of-Speech Tagging

- **Definition:** Part-of-speech (POS) tagging is the process of assigning word classes or tags (such as noun, verb, adjective, etc.) to each word in a sentence.
- **Purpose in NLP:** Helps in syntactic analysis, information extraction, and improving the performance of downstream NLP tasks (e.g., named entity recognition, parsing, etc.)
- **Importance:** Helps the model understand the structure of the sentence.

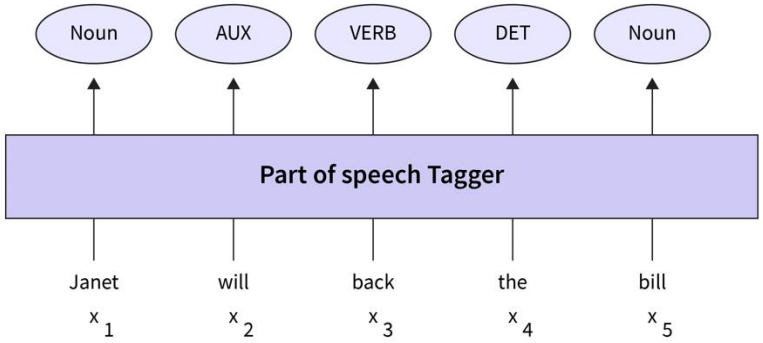


Image: https://shorturl.at/aBaxg

Why POS Tagging is useful

- *POS tagging is often used as an early step in an NLP pipeline.*
- *Text-to-speech e.g., **How do you pronounce "read"?***
- *If you know the tag, you can back off to it in other tasks. Using proper nouns is a good back off method for finding people.*

E.g. "Alice loves reading books. She always has a pile of them on her desk."

  - *Question: What does the word "She" refer to?*
  - *POS Tagging:*
    - *Alice → Proper Noun*
    - *She → Pronoun*
    - *her → Pronoun*
  - *Using the information from POS tags, we can back off to the rule that pronouns (like "She" and "her") typically refer to a previously mentioned proper noun (in this case, "Alice"). This allows us to resolve that "She" and "her" refer to "Alice."*

Do we need POS Tagging while we have LLMs?

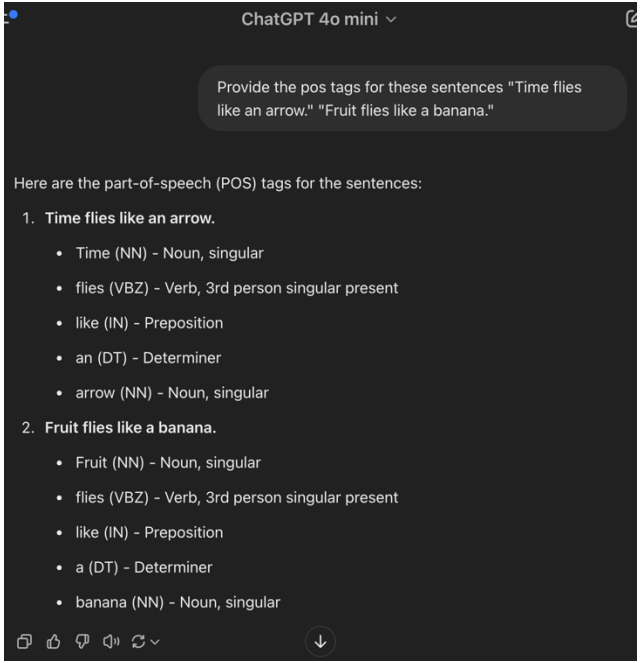
With the rise of **LLMs**, many NLP tasks, including POS tagging, can be performed automatically.

However, **understanding POS tagging remains crucial** for multiple reasons.

Why is POS tagging still an important technique in the age of LLMs?

Why do we need POS Tagging when LLMs exist?

- LLMs **do not replace** foundational NLP concepts - they build upon them. POS tagging is a fundamental linguistic task that helps in syntactic analysis, semantic role labelling, named-entity recognition (NER), and many downstream tasks, such as machine translation and information retrieval.
- LLMs are not perfect – they can make mistakes. See picture on the right.
- POS tagging allows us to generate tags independently and accurately (or to detect and correct the errors made by LLMs).



Categories of POS

Open and closed classes

- Closed classes have a fixed membership of words: determiners, pronouns, prepositions
- Closed class words are usually function word: frequently occurring, grammatically important, often short (e.g. of, it, the, in)
- Open classes: nouns, verbs, adjectives and adverbs (allow new addition of word)

Open class (lexical) words

Nouns

Proper

IBM
Italy

Common

cat / cats
snow

Verbs

Main

see
registered

Adjectives *old older oldest*

Adverbs *slowly*

Numbers

122,312
one

… more

Closed class (functional)

Determiners *the some*

Conjunctions *and or*

Pronouns *he its*

Modals

can
had

Prepositions *to with*

Particles *off up*

Interjections *Ow Eh*

… more

POS Tagging

Words often have more than one POS: *back*

- The *back* door = adjective
- On my *back* = noun
- Win the voters *back* = adverb
- Promised to *back* the bill = verb

The POS tagging problem is to determine the POS tag for a particular instance of a word.

Common POS Tags

Nouns (NN, NNS, NNP, NNPS): Person, places, things

Verbs (VB, VBD, VBG, VBN, VBP, VBZ): Actions or states of being

Adjectives (JJ, JJR, JJS): Words that describe nouns

Adverbs (RB, RBR, RBS): Words that modify verbs, adjectives, or other adverbs

Pronouns (PRP, PRP$, WP, WP$): Words that take the place of nouns

Prepositions (IN): Words that show relationships between nouns or pronouns

Conjunctions (CC): Words that connect phrases, clauses, or sentences

Penn Treebank POS Tags

Tag	Description	Example	Tag	Description	Example	Tag	Description	Example
CC	coordinating conjunction	*and, but, or*	PDT	predeterminer	*all, both*	VBP	verb non-3sg present	*eat*
CD	cardinal number	*one, two*	POS	possessive ending	*’s*	VBZ	verb 3sg pres	*eats*
DT	determiner	*a, the*	PRP	personal pronoun	*I, you, he*	WDT	wh-determ.	*which, that*
EX	existential ‘there’	*there*	PRP$	possess. pronoun	*your, one’s*	WP	wh-pronoun	*what, who*
FW	foreign word	*mea culpa*	RB	adverb	*quickly*	WP$	wh-possess.	*whose*
IN	preposition/ subordin-conj	*of, in, by*	RBR	comparative adverb	*faster*	WRB	wh-adverb	*how, where*
JJ	adjective	*yellow*	RBS	superlatv. adverb	*fastest*	$	dollar sign	$
JJR	comparative adj	*bigger*	RP	particle	*up, off*	#	pound sign	#
JJS	superlative adj	*wildest*	SYM	symbol	+,%, &	“	left quote	‘ or “
LS	list item marker	*1, 2, One*	TO	“to”	*to*	”	right quote	’ or ”
MD	modal	*can, should*	UH	interjection	*ah, oops*	(	left paren	[, (, {, <
NN	sing or mass noun	*llama*	VB	verb base form	*eat*	)	right paren	],), }, >
NNS	noun, plural	*llamas*	VBD	verb past tense	*ate*	,	comma	,
NNP	proper noun, sing.	*IBM*	VBG	verb gerund	*eating*	.	sent-end punc	. ! ?
NNPS	proper noun, plu.	*Carolinas*	VBN	verb past part.	*eaten*	:	sent-mid punc	: ; ... – -

See: https://bit.ly/39mKCp0

Use as a reference

Two Methods for POS Tagging

- Rule-based tagging

  ENGTWOL (English Treebank Word and Lemma)

- Machine Learning methods

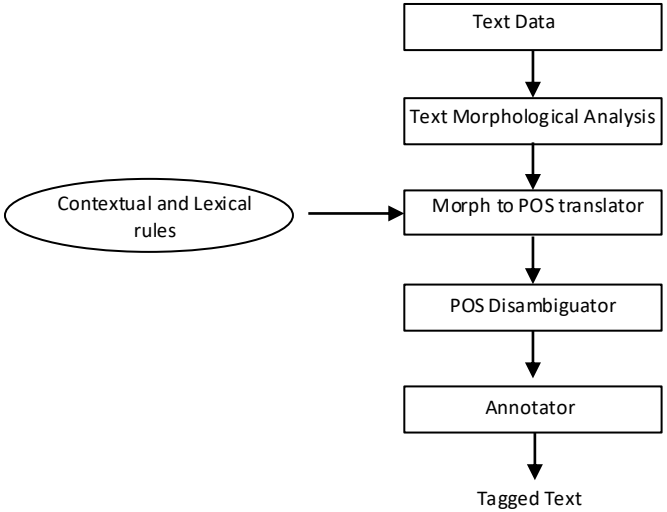
  Probabilistic sequence models

  HMM (Hidden Markov Model) tagging

  Neural Network based approach

Rule-Based Tagging

- ENGTWOL (English Treebank Word and Lemma)
  - This tagger are knowledge-driven tagger.
  - The rules in Rule-based POS tagging are built manually.
  - The information is coded in the form of rules.
  - We have some limited number of rules approximately around 1000.
  - It is a specialized POS tag set used to annotate English text with both POS tags and lemmas.
  - Lemmas: The canonical or dictionary form of a word (e.g., "running" → "run").



Machine Learning based POS Tagging

- Probabilistic sequence models
  - The probabilistic model for tagging assumes sentences are generated probabilistically.
  - Sentence generation involves selecting tags and words based on conditional probabilities.
  - Tags are chosen based on the previous tag with a probability
  - The model is a bigram tag model where each tag depends only on the previous tag.

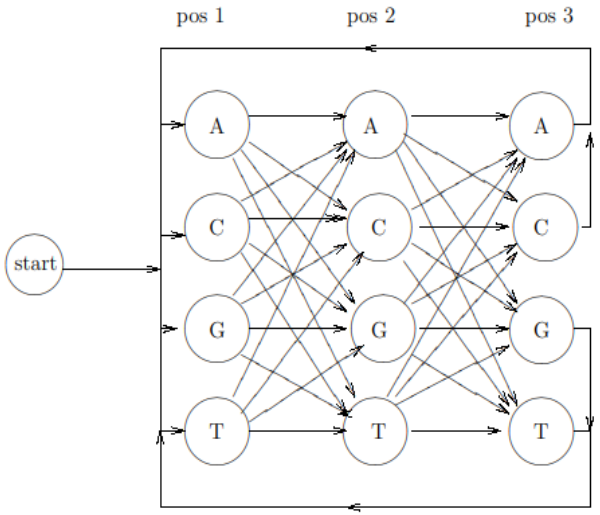


Fig: Probabilistic sequence models.

Machine Learning based POS Tagging

- HMM (Hidden Markov Model) tagging
  - An HMM models a system as a sequence of hidden states.
  - At each time step, the HMM emits an observable symbol or observation based on the current hidden state.
  - HMMs incorporate transition probabilities, which specify the likelihood of moving from one hidden state to another at each time step.
  - For each hidden state, there are associated emission probabilities.

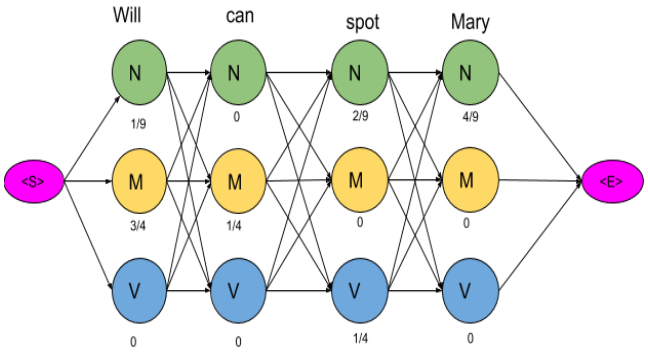
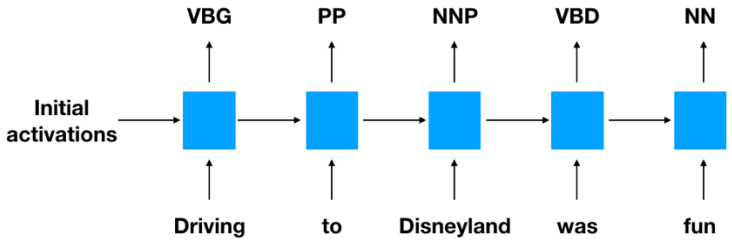


Fig: Hidden Markov Model (HMM).

Neural Network based POS Tagging

- Neural Network-based tagging
  - Input: A sequence of words or tokens.
  - Embedding Layer: Converts words into dense vectors.
  - RNN/LSTM Layers: Sequence models for capturing dependencies across words.
  - Output Layer: Softmax layer to predict the POS tag for each word in the sequence.



How difficult is POS tagging?

About 11% of the word types in the Brown corpus are ambiguous with regard to part of speech

But they tend to be very common words. E.g., *that*

- I know *that* he is honest = IN (Subordinating conjunction)
- Yes, *that* play was nice = DT (Determiner)
- Yes, *that* was nice = PRP (Pronoun)
- You can't go *that* far = RB (Adverb)

40% of the word tokens are ambiguous

Sources of information

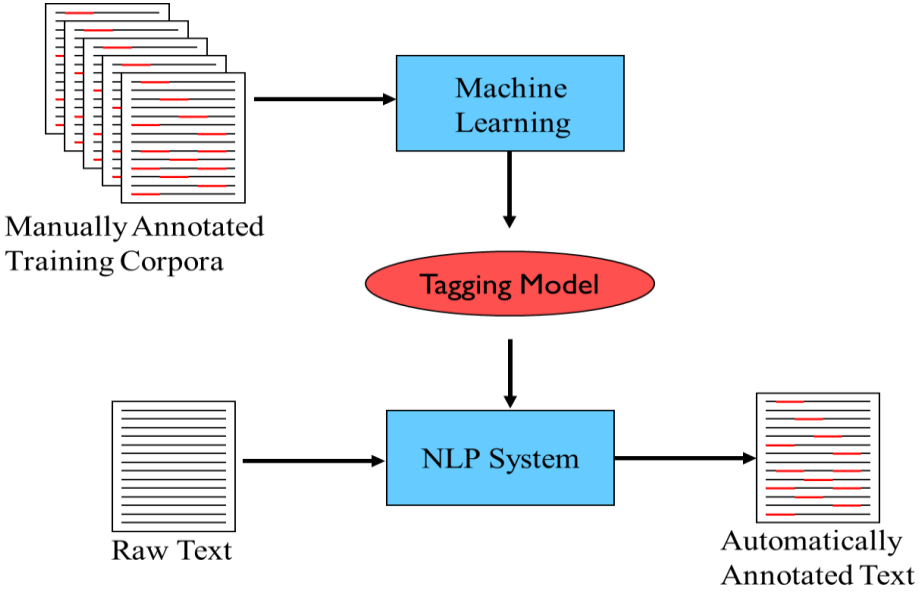
What are the main sources of information for POS tagging?

- Knowledge of neighboring words
  - Bill saw that man yesterday
  - NNP NN DT NN NN
  - VB VB IN VB NN
- Knowledge of word probabilities
  - *man* is rarely used as a verb….

The latter proves the most useful, but the former also helps

POS Tags
NNP: Proper Noun
NN: Noun
DT: Determiner
VB: Verb
IN: Preposition or Subordinating Conjunction

Learning Approach



More and Better Features ➔ Feature-based tagger

Can do surprisingly well just looking at a word by itself:

- Word the: the → DT
- Lowercased word Importantly: importantly → RB
- Prefixes unfathomable: un- → JJ
- Suffixes Importantly: -ly → RB
- Word shapes 35-year: d-x → JJ

POS Tags
DT: Determiner
RB: Adverb
JJ: Adjective

These can all become "features" in a machine learning model for POS tagging.

How to improve supervised results?

Build better features!

POS Tags
PRP: Pronoun
VBD: Verb, Past Tense
IN: Subordinating Conjunction
RB: Adverb
JJ: Adjective
NNP: Noun, Plural
VBN: Verb, Past Participle

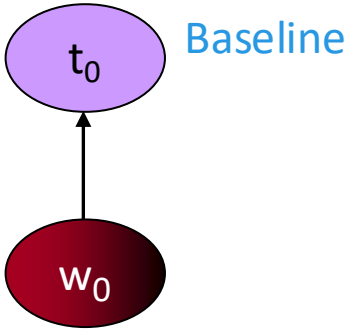
		IN					
PRP	VBD	~~RB~~	RB	IN	PRP	VBD	.
They	left	as	soon	as	he	arrived	.

- We could fix this with a feature that looked at the next word

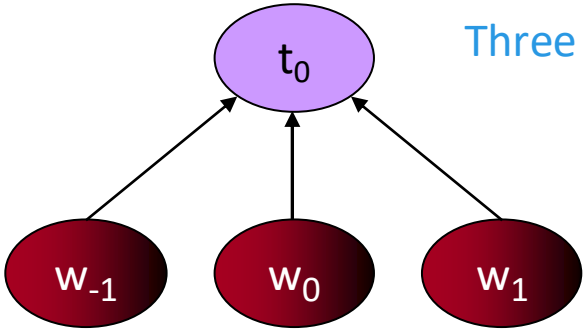
JJ					
~~NNP~~	NNS	VBD		VBN	.
Intrinsic	flaws	remained		undetected	.

- We could fix this by linking capitalized words to their lowercase versions

Tagging Without Sequence Information



Baseline



Three Words

Model	Features	Token	Unknown	Sentence
Baseline	56,805	**93.69%**	82.61%	26.74%
3Words	239,767	**96.57%**	86.78%	48.27%

Out-of-vocabulary (OOV) words in POS Tagging

Even high-accuracy POS taggers fail when:

- Applied to a new domain
- Encountering out-of-vocabulary (OOV) words
- Processing informal or technical language

High accuracy on Penn Treebank ≠ robustness in the real world.

What if the training and testing distributions are different?

- A model is trained on one distribution P_train(X,Y). But tested on another P_test(X,Y)
- When P_train ≠ P_test → performance drops.

Example

- Training sentence: “The company reported earnings yesterday.”
- Twitter (i.e., X) sentence: “bro that movie lowkey slaps fr 💀”
- The tagger struggles because it has never seen:
  - "lowkey"
  - "slaps" (slang sense)
  - emoji

Solution Strategy: Better Features

- Instead of memorising words, we design features that generalise.
- Learn patterns, not tokens.
- Potential useful features:
  - Character n-grams
  - Suffix & Prefix features
  - Word Shape
  - Context Expansion

Feature 1: Character n-grams

- An **n-gram** is a contiguous sequence of **n items** from a text. The “items” can be characters, words or subwords.

- Example: for 'bioinformatics’, character n-grams can be:
  - bio
  - info
  - matics
  - -ics

- Captures morphology and domain signals.

Feature 2: Suffix & Prefix Features

Examples:

- -ly → RB
- -ed → VBD/VBN
- -tion → NN
- -ase → enzyme (NN)
- -itis → disease (NN)

Works for unseen words.

Feature 3: Word Shape

Word → Shape

BRCA1 → XXXX d

2026 → dddd

iPhone12 → xXxxxxdd

USA → XXX

Helps detect proper nouns and technical terms.

Feature 4: Context Expansion

Add contextual features:

- Previous two tags

- Next word

- Surrounding POS

Context reduces reliance on memorised words.

Why Feature-Based Solutions Work

They:

✓ Capture morphology

✓ Reduce dependence on fixed vocabulary

✓ Improve OOV handling

✓ Adapt better across domains

Move from word memorisation → pattern learning.

We will try some of these features in the workshop coding exercise.

Discussion

If deploying a POS tagger on TikTok comments:

What new features would you design?

- Emoji feature?
- Hashtag detection?
- Elongated words ('soooo')?
- URL detection?

Applications of POS Tagging

Information Retrieval & Search

POS tags help distinguish between different senses of a word.

Example:

- Query: "fast cars"
- POS tagging helps identify "fast" as an adjective modifying "cars" (noun), leading to more accurate results compared to simply searching for "fast" and "cars" separately.

Sentiment Analysis

POS tags help identify adjectives and adverbs that contribute to sentiment.

Example:

- "The service was consistently disappointing. The food was surprisingly delicious."
- POS tagging identifies "disappointing" and "delicious" as sentiment-bearing adjectives. A sentiment analysis system can then determine that the service had a negative sentiment and the food had a positive sentiment, even within the same review.

Applications of POS Tagging (cont’)

- **Text to Speech Systems**
  - In Text-to-Speech systems, POS tagging is crucial for determining the correct pronunciation, intonation, and stress patterns based on the role of words in a sentence..
  - Example:
    - Query: "I will read the book.” vs “I have read the book”.
    - POS tagging helps identify that the word “read” in the first sentence is a verb in the base form (VB), and the second one is in the past tense (VBD), so the systems would pronounce it differently.

Questions?